
Billion Dollar Disasters in the Contiguous United States: Linking Climate Variability and Income Across States (2015-2023)

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Introduction and Background

Over the past several decades, the United States has experienced an increase in frequency and cost of extreme weather and climate events. These events, classified by the National Oceanic and Atmospheric Administration (NOAA) as “billion-dollar disasters”, include hurricanes, droughts, wildfires, floods, and severe storms that each result in at least one billion dollars in damage. Since 2015, the cumulative cost of these disasters has increased drastically driven by a combination of factors including change in climate, increasing exposure of infrastructure, and population growth in vulnerable regions. As these events become more frequent, and more costly, understanding the environmental and socioeconomic conditions associated with them has become increasingly important.

These billion dollar disasters are not evenly distributed across the United States, nor are their impacts uniform. Regional differences in weather and climate play a critical role in shaping the types and frequency of extreme events that occur. Variations of temperature and precipitation, for example, influence the likelihood of hazards, such as drought, wildfire, and flooding. Warmer temperatures can intensify evaporation and soil drying, increasing drought severity and wildfire risk. Additionally, changes in precipitation patterns can lead to more intense rainfall events and flooding. As a result, examining disaster patterns alongside climate variability provides important context for understanding why certain states experience more frequent or more costly events than others.

In addition to environmental drivers, socioeconomic factors, such as income, also shape how disasters are experienced and managed. States with higher average income may have greater resources for disaster preparation, infrastructure resilience, and recovery efforts, while lower-income regions may face challenges responding to and rebuilding after extreme events. By integrating state-level disaster data with temperature, precipitation, and income, this project aims to provide a more comprehensive analysis of billion dollar disasters across the United States.

This project seeks to answer several key research questions related to the spatial distribution, climatic drivers, and socioeconomic context of billion dollar disasters in the United States:

1. **Which state experienced the most expensive disaster, and what type was it?** To examine this question, we will examine the cost of individual disasters and identify the maximum recorded loss. We will also classify the disaster type (e.g. flood, drought, wildfire, hurricane) to better understand which type of disaster is associated with the greatest economic impacts.
2. **How do the average income levels of the five states with the most expensive disasters compare to the national average?** This question introduces a socioeconomic perspective by evaluating whether the most disaster-prone states differ economically from the rest of the country.
3. **How do yearly temperature anomalies relate to the most expensive billion dollar disasters across the United States?** This question explores the potential relationship between long-term warming trends and disaster frequency. By comparing annual temperature anomalies to the number of billion-dollar disasters each year, we can assess whether years with above-average temperatures are associated with increased disaster occurrence. This helps evaluate whether rising temperatures may be linked to heightened climate-related risk.
4. **How do yearly precipitation anomalies relate to the most expensive of billion dollar disasters across the United States?** This question examines how deviations in precipitation from normal conditions influence disaster frequency. By analyzing annual precipitation anomalies alongside disaster counts, we can determine whether unusually wet or dry years correspond to more frequent billion-dollar disasters. This provides insight into how hydrologic, such as droughts or extreme rainfall, may contribute to economic losses.

Conceptual Diagrams:

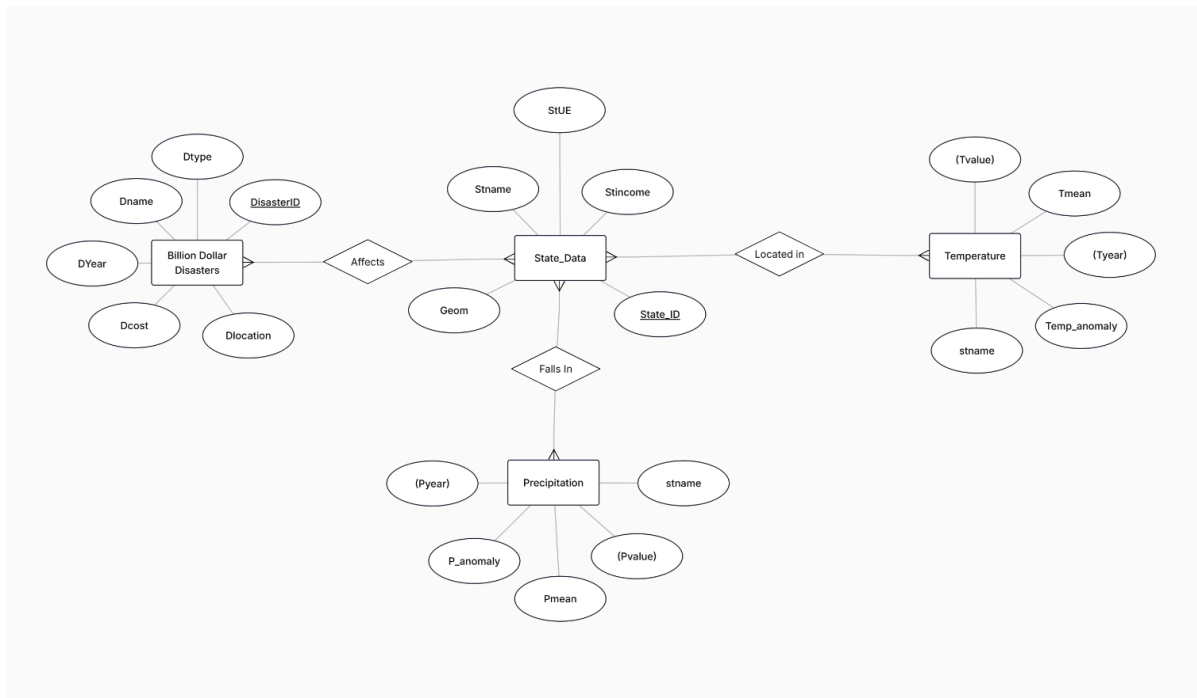


Figure 1: An ER diagram showing the relationship between the spatial and nonspatial objects and their attributes

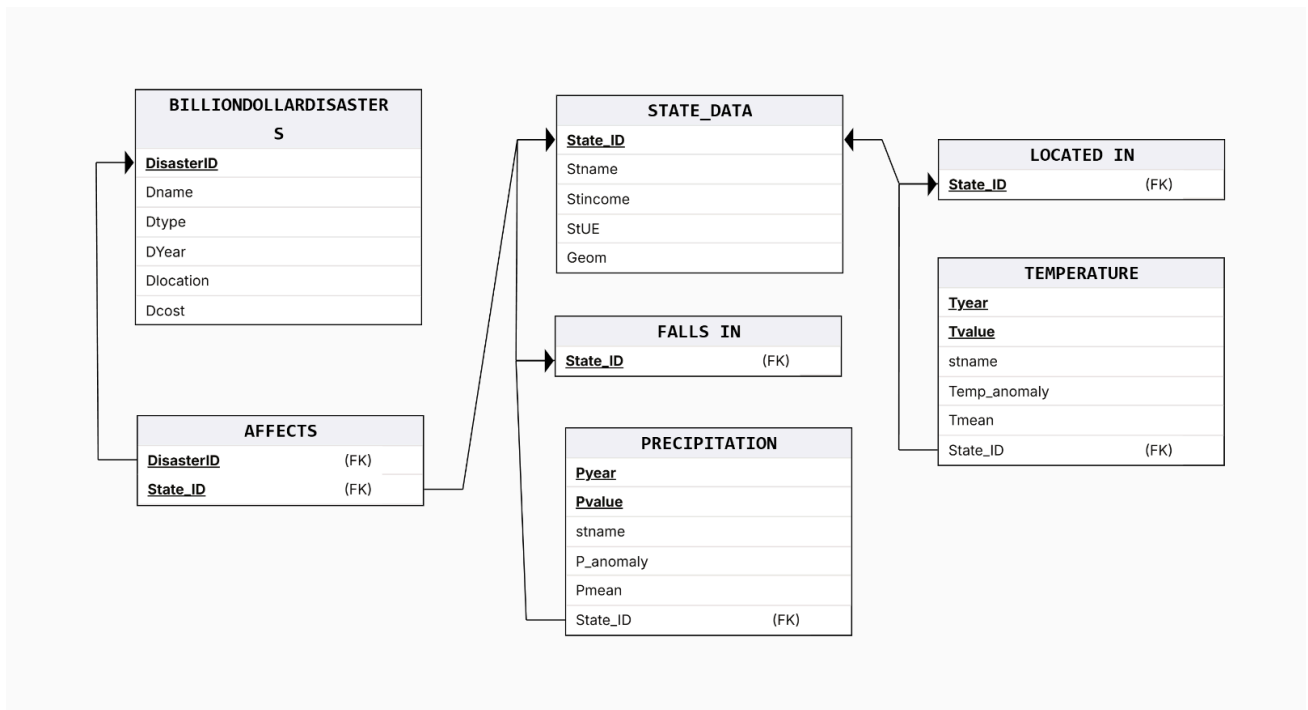


Figure 2: A relational schema showing the logical design of the spatial database (the arrows originate from the foreign key of each entity and end at the entities with the primary keys)

Methods:

Database Requirements

The database for this project needed the compatibility of analysis and data storage. This includes a comprehensive database containing the values of precipitation, temperature, income, and billion dollar disasters on a state level. Using these values the concept of our project aimed to determine the interconnection of economics and climate.

Datasets

We have four nonspatial datasets and one spatial. The spatial dataset is from NOAA and solely provides the polygon information for each state. We reference this dataset in almost every query because it is important to know spatially where each state has been impacted. The other datasets show precipitation (NOAA), temperature (NOAA), billion dollar disasters (NOAA) and unemployment rate (USDA). These are all credible datasets as they had been created and funded by the federal government. The data in each of these layers are separated by state and year to make analyzing the impact of natural disasters more manageable.

Entity-Relationship Model

To begin, the focus was aimed at creating a map of our individual variables and their features. This would aim to support storage, querying, and analysis for our database. State information forms a spatial anchor for all the data used within the database. The state layer contains spatial information like the feature type, latitude and longitude, and coordinate system. We combined this layer with state unemployment data via an outer join. The other three layers: billion disasters, temperature, and precipitation are all separate entities connected to the state_id primary key. The model shows the relationship between the four entities and their maximum cardinalities. Each of the entities have a many to many relationship because one temperature can be the same in multiple states and one state can have many temperatures. This same logic applies to disasters and precipitation.

Results:

- Which state experienced the most expensive disaster, and what type was it?

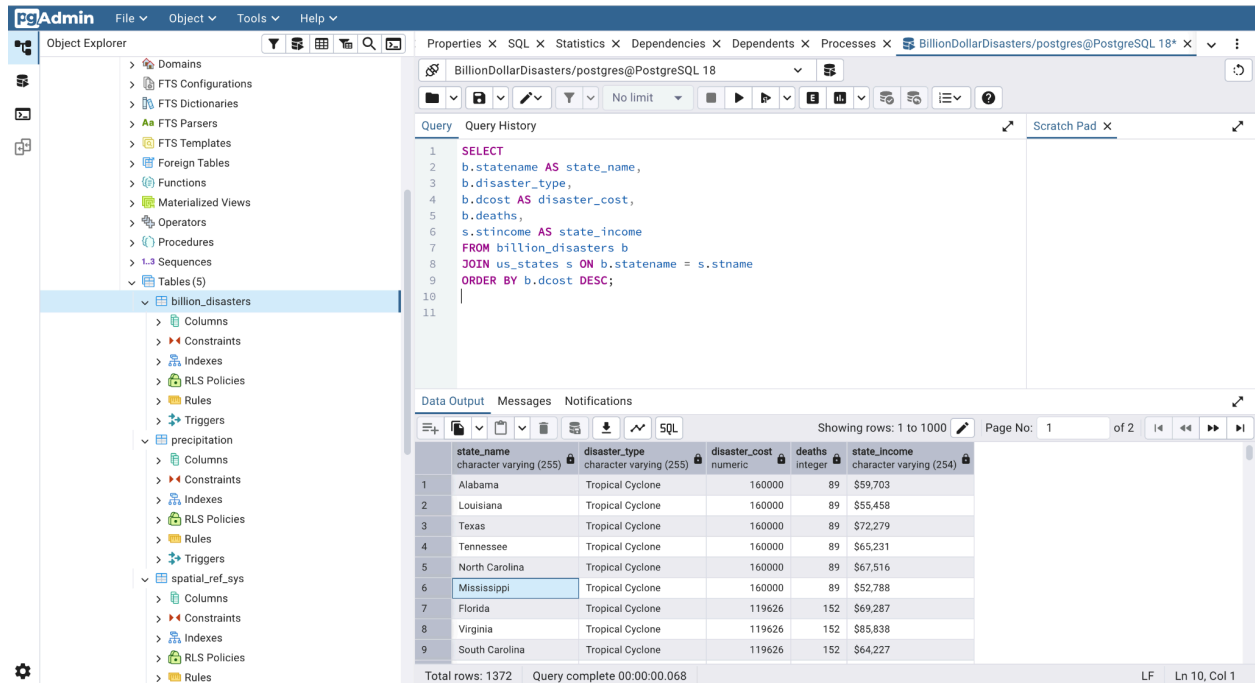


Figure 3: Query results determining most expensive disaster and corresponding states.

```
SELECT
b.statename AS state_name,
b.disaster_type,
b.dcost AS disaster_cost,
b.deaths,
s.stincome AS state_income
FROM billion_disasters b
JOIN us_states s ON b.statename = s.stname
ORDER BY b.dcost DESC;
```

The findings were that it was not a singular state impacted by the most expensive disaster, but instead a region. The most expensive disaster is a tropical cyclone. Instead of one state impacted

by the tropical cyclone it was Tennessee, Mississippi, Texas, Alabama, Louisiana, North Carolina. They suffered the cost of \$16000.

- **How do the average income levels of the five states with the most expensive disasters compare to the national average?**

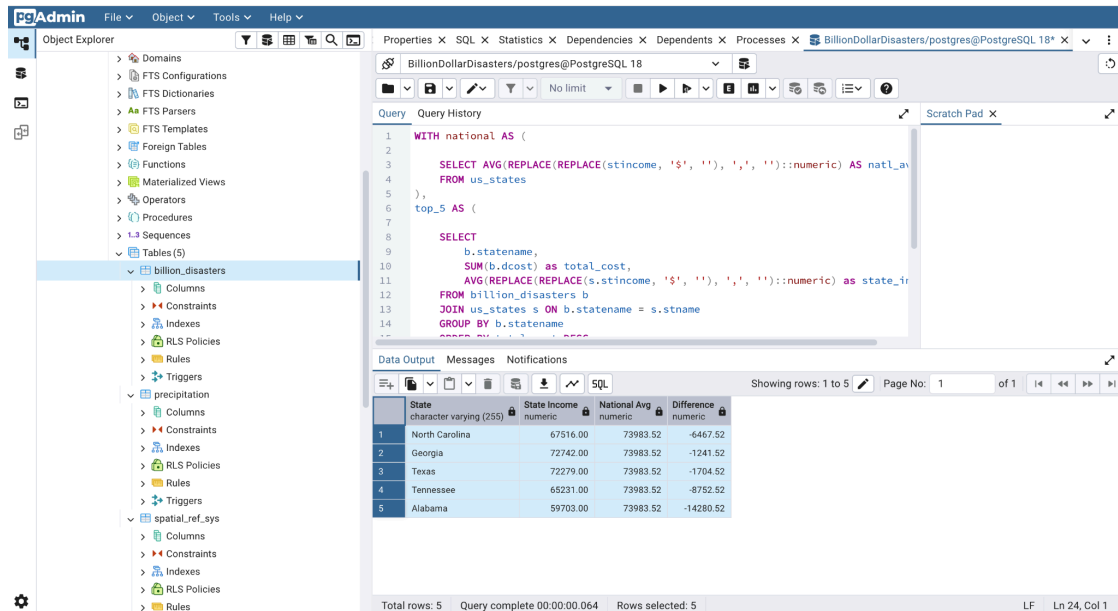


Figure 4: Query results identifying national income average, state income, and the difference between for states most impacted by disasters economically.

Query:

WITH national AS (

SELECT AVG(REPLACE(REPLACE(stincome, '\$', ''), ',', ''))::numeric) AS natl_avg

FROM us_states

),

top_5 AS (

SELECT

b.state_name,

SUM(b.dcost) as total_cost,

AVG(REPLACE(REPLACE(s.stincome, '\$', ''), ',', ''))::numeric) as state_income

FROM billion_disasters b

```
JOIN us_states s ON b.statename = s.stname
GROUP BY b.statename
ORDER BY total_cost DESC
LIMIT 5
)
SELECT
    t.statename AS "State",
    ROUND(t.state_income, 2) AS "State Income",
    ROUND(n.natl_avg, 2) AS "National Avg",
    ROUND(t.state_income - n.natl_avg, 2) AS "Difference"
FROM top_5 t, national n;
```

The resulting metrics of the state income contrasted against the state average income yielded a trend. The state incomes all fell below national average income by a factor of at least \$1200. This means that the most expensive disaster impacted a region where the state income fell lower on the end of the economy.

- **How do average temperature anomalies relate to the most expensive billion dollar disasters across the United States?**

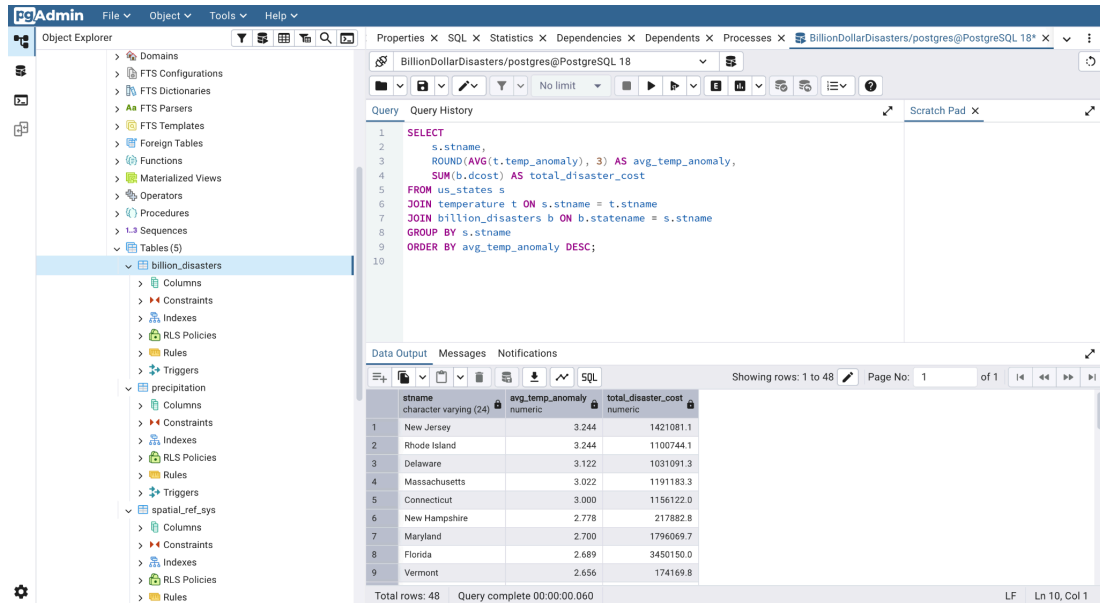


Figure 5: Query identifying average temperature anomalies in regions affected by disasters.

```

SELECT
    s.stname,
    ROUND(AVG(t.temp_anomaly), 3) AS avg_temp_anomaly,
    SUM(b.dcost) AS total_disaster_cost
FROM us_states s
JOIN temperature t ON s.stname = t.stname
JOIN billion_disasters b ON b.statename = s.stname
GROUP BY s.stname
ORDER BY avg_temp_anomaly DESC;

```

Determining the relationship between average temperature anomalies and expensive disasters yielded a linear relationship between the values. The more expensive that a disaster was, linked it with a higher temperature anomaly in the region. In areas where there was a lower disaster cost, the chance of a temperature anomaly lowered.

- **How do average precipitation anomalies relate to the most expensive of billion dollar disasters across the United States?**

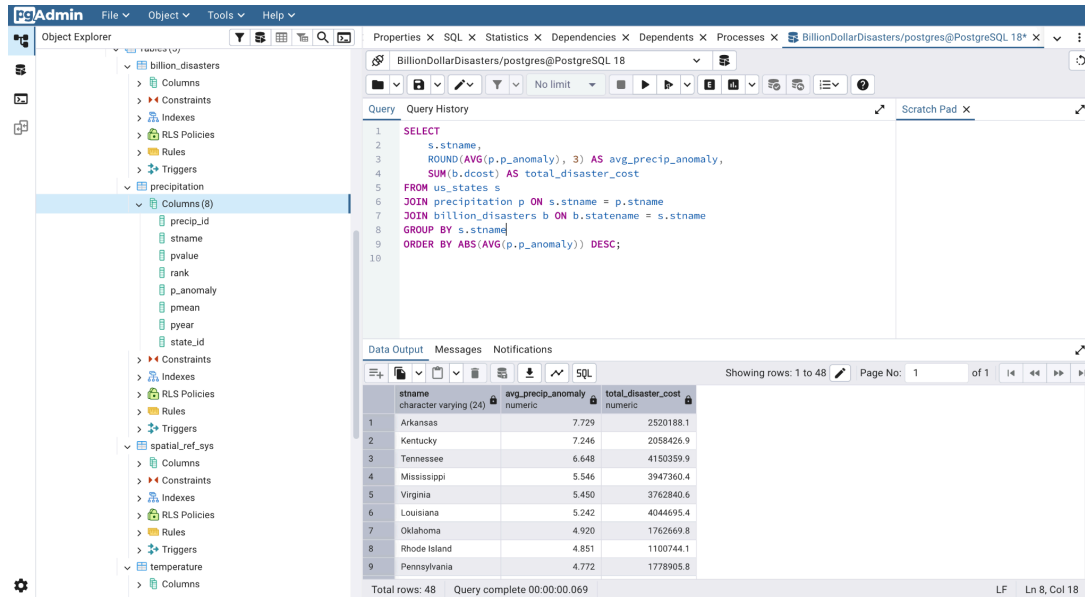


Figure 6: Query identifying average precipitation anomalies in regions affected by disasters.

```

SELECT
    s.stname,
    ROUND(AVG(p.p_anomaly), 3) AS avg_precip_anomaly,
    SUM(b.dcost) AS total_disaster_cost
FROM us_states s
JOIN precipitation p ON s.stname = p.stname
JOIN billion_disasters b ON b.statename = s.stname
GROUP BY s.stname
ORDER BY ABS(AVG(p.p_anomaly)) DESC;

```

The relationship between precipitation anomalies and the most expensive of billion dollar disasters followed a similar trend to the temperature anomalies and billion dollar disasters. The relationship remains semi-linear with a low standard deviation. Overall, the more expensive the disaster, the higher the chance of an extreme precipitation anomaly value. The highest value being 7.729.

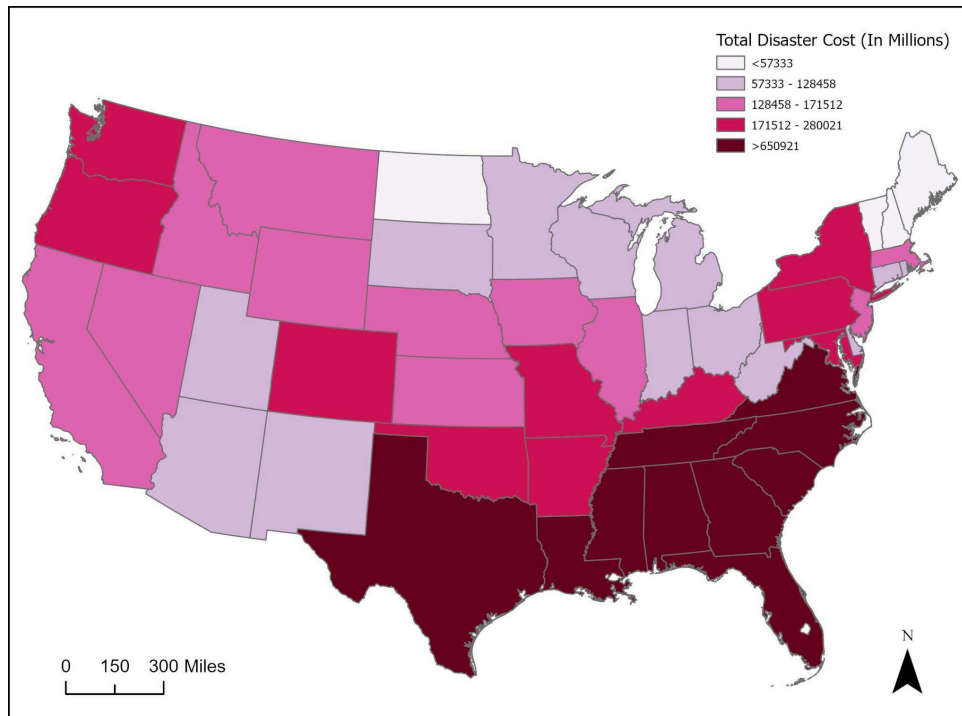


Figure 7: Map of the contiguous United States showing the total cost of billion dollar disasters per state

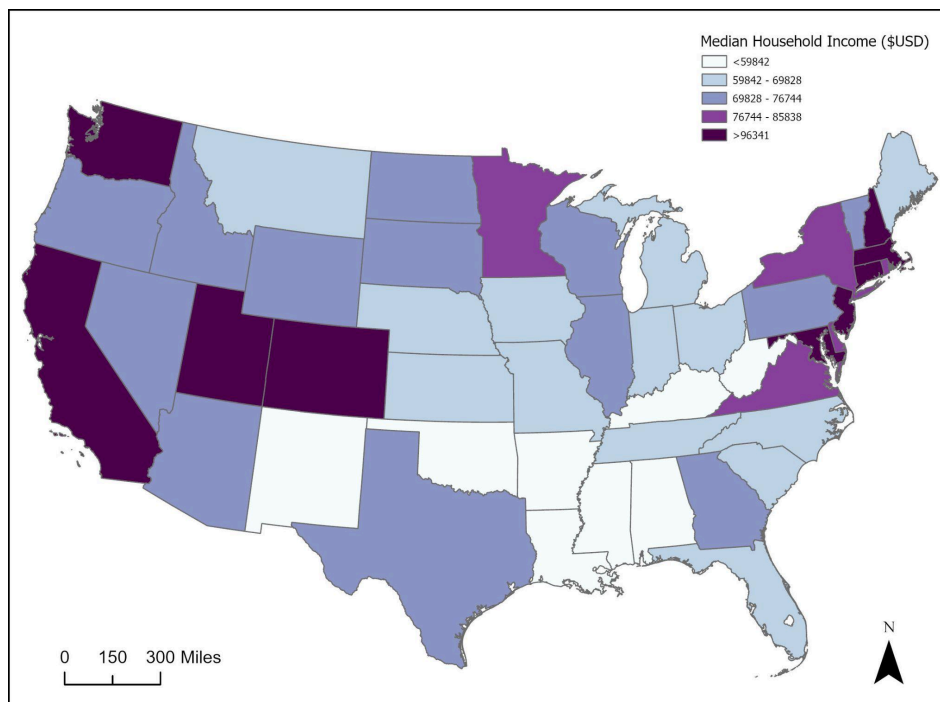


Figure 8: Map of the contiguous United States showing the distribution of median income

References

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[https://www.ncei.noaa.gov/access/billions/events/WY/2015-2023/?disasters\[\]=all-disasters](https://www.ncei.noaa.gov/access/billions/events/WY/2015-2023/?disasters[]=all-disasters)

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